Preliminary aerodynamic estimate

1 Purpose and scope

This note turns the dimensions and masses proposed for ALIMOCHE into a first drag polar and into estimates of stall, glide, sink, thrust and power. It serves to compare configurations in P01. The numbers are preliminary-design assumptions, not authorised speeds nor a demonstration of safety or compliance.

The main case is ASM-01 of the specification [2]: Lillo at 40 °C, with density 1.04 kg/m³. ISA at sea level is included for comparison. The SG 38 three-view drawing is a historical visual reference; the calculations correspond to the rectangular wing proposed for ALIMOCHE.

2 Input data and first-order results

Table 1 gives the input data.

Table 1. Input data.

Quantity	Value	Origin or interpretation
Span b	10.0 m	Working value from the project lead
Mean chord c	1.0 m	Assumed constant along the whole wing
Wing area S	10.0 m²	$S = bc$; excludes stabilisers
Aspect ratio AR	10.0	$\mathrm{AR} = b^2/S$
Target empty mass	70 kg	ALM-00 requires less than 70 kg [1]
Maximum equipped pilot	90 kg	M3 of the specification [2]
Maximum calculation mass	160 kg	70 + 90 kg proposal, not an approved MTOW
Calculation weight	1569 N	mg, with $g = 9.80665$ m/s²
Maximum wing loading	16.0 kg/m²	Equivalent to 157 N/m²

With 70 kg of empty mass and 90 kg of pilot, the 160 kg are used up. There is no mass margin for instruments, ballast, protection or construction changes. In addition, the specification sets less than 70 kg, so the 70 kg value used here is a conservative limit case for performance. A mass budget is required before setting the MTOW.

The specification also considers a 50 kg pilot and a nominal case of 75 kg. If the empty mass were 70 kg, total masses of 120 and 145 kg would result, respectively, before any ballast. The light case will need a separate centre-of-gravity analysis.

3 Provisional aerodynamic model

$C_{L,\max} = 1.50$ is taken as the central case. It is plausible for an initial exploration, but may prove optimistic for the actual aerofoil, Reynolds number, construction and three-dimensional stall. The range 1.20–1.50 is also calculated. This value is not assigned as a requirement or as a demonstrated property.

For an open structure inspired by the SG 38, $C_{D,0} = 0.055$ is used, referred to the 10 m² of wing. It includes, in aggregate, wing, exposed pilot, structure and bracing; it does not come from a measurement of ALIMOCHE. $e = 0.70$ is adopted for the rectangular wing, from which $k = 1/(\pi\ \mathrm{AR}\ e) = 0.0455$. This pair of values gives a maximum glide ratio close to 10, of the order of the SG 38 preserved at the Deutsches Museum [6]. The SG 38 coefficients have not been transferred directly: its area and configuration are different. The model is given by Eq. 1 and Eq. 2 [4], and the resulting polar is shown in Figure 1.

$$q = \frac{1}{2}\rho V^2, \quad C_L = \frac{W}{qS}, \quad C_{D,i} = kC_L^2, \quad C_D = C_{D,0} + C_{D,i} \tag{1}$$

$$D = qSC_D, \quad P_{\mathrm{aero}} = DV, \quad v_{\mathrm{sink}} \approx \frac{P_{\mathrm{aero}}}{W} \tag{2}$$

where q is the dynamic pressure, ρ the air density, V the true airspeed, W the weight, D the drag, P_{aero} the aerodynamic power and v_{sink} the sink rate.

The parabolic polar is used only before the stall and in near-steady glide. Close to $C_{L,\max}$, the model no longer describes faithfully the rise in drag or the changes in control. Power and climb thrust are presented as hypothetical aerodynamic loads to study a future powerplant; the basic glider has no engine.

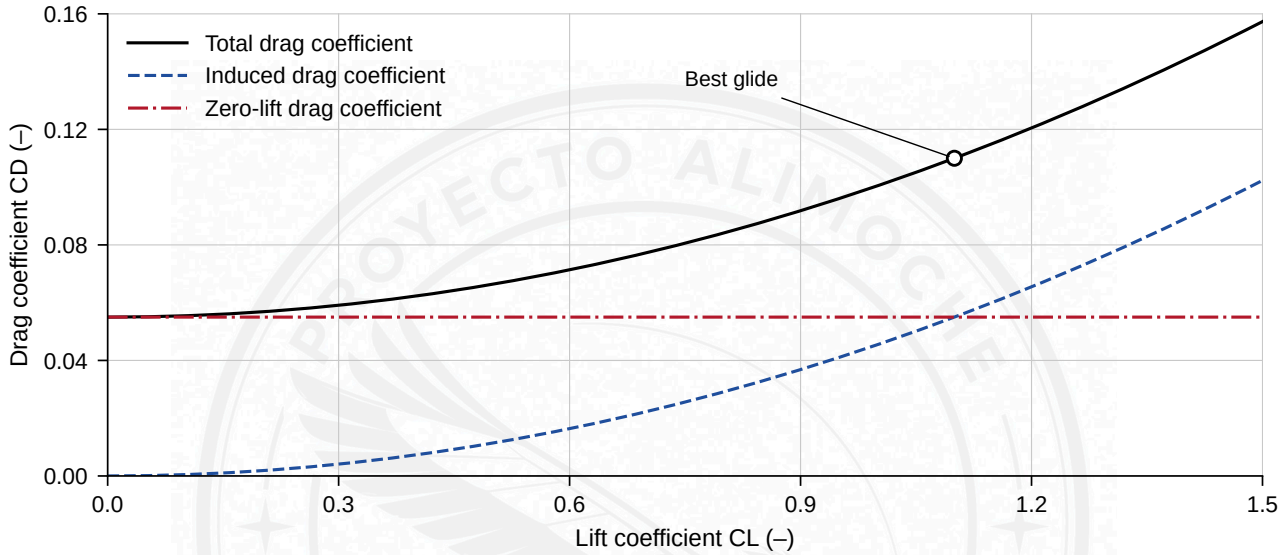


Figure 1. Provisional drag polar of the complete aircraft, $C_D = C_{D,0} + kC_L^2$ with $C_{D,0} = 0.055$ and $k = 0.0455$. The circle marks the best-glide point of the model; the model applies before the stall.

4 Estimated glide performance

Table 2 gives the glide performance at 160 kg and Figure 2 the sink-rate polar [5].

Table 2. Estimated glide performance.

Quantity at 160 kg	ISA sea level	Lillo 40 °C
Density	1.225 kg/m³	1.04 kg/m³
Stall speed with CLmax 1.50	47.0 km/h	51.1 km/h
Stall speed if CLmax were 1.20	52.6 km/h	57.1 km/h
Best-glide speed	54.9 km/h	59.6 km/h
Maximum glide ratio	10.0	10.0
Drag at best glide	157 N	157 N
Sink rate at best glide	1.53 m/s	1.66 m/s

The best-glide speed lies only 17 % above the stall speed calculated with $C_{L,\max} = 1.50$. This figure must therefore not be turned directly into an operating speed. The minimum-sink speed of the mathematical model lies practically at the edge of the stall; it is not published as usable performance until a real polar and handling margins are obtained.

With a nominal 75 kg pilot and a total mass of 145 kg, the calculated stall speed would be 44.8 km/h in ISA and 48.6 km/h at Lillo; with a 50 kg pilot, 40.7 and 44.2 km/h. These values only reflect the effect of mass with unchanged geometry and $C_{L,\max}$.

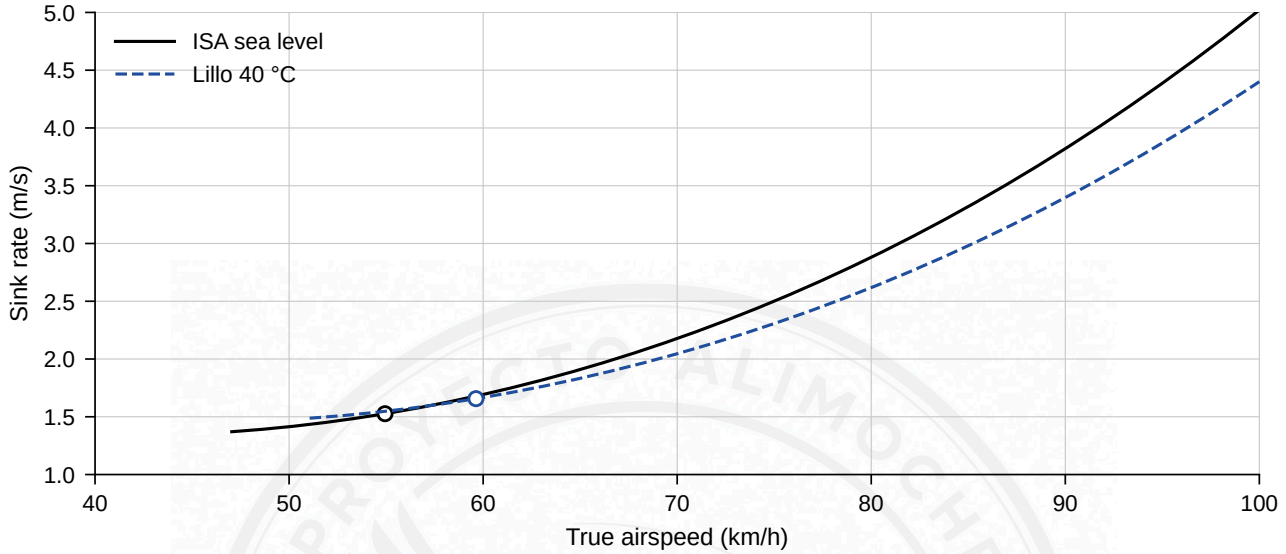


Figure 2. Sink-rate polar in still air, in true airspeed, at 160 kg. Each curve starts at the calculated stall speed for its atmosphere; the circles mark best glide.

A glide ratio of 10 implies, at the optimum point and with no wind, a loss of about 100 m for every 1 km of horizontal distance. Bungee launch distance, height reached and landing run have not been calculated: the launch system, the available energy, the terrain and the coefficients near the stall are missing.

5 Drag, thrust and power

In unpowered glide, the weight component balances the drag. For a possible propulsion system, the equilibrium thrust in level flight would be approximately $T = D$, and for a steady climb at rate R it is estimated as $T \approx D + WR/V$. The aerodynamic power would be $DV + WR$. With an assumed propulsive efficiency of 0.70, the shaft power is that sum divided by 0.70. Electrical losses, installation drag, mass variations and take-off are not included. Table 3 gives the values at best-glide speed, and Figure 3 and Figure 4 their variation with speed.

Table 3. Hypothetical thrust and power at best-glide speed, Lillo, 160 kg.

Lillo at 59.6 km/h	Level flight	Climb 1 m/s	Climb 2 m/s
Thrust required	157 N	252 N	346 N
Aerodynamic power	2.60 kW	4.17 kW	5.74 kW
Shaft power, η = 0.70	3.71 kW	5.96 kW	8.20 kW

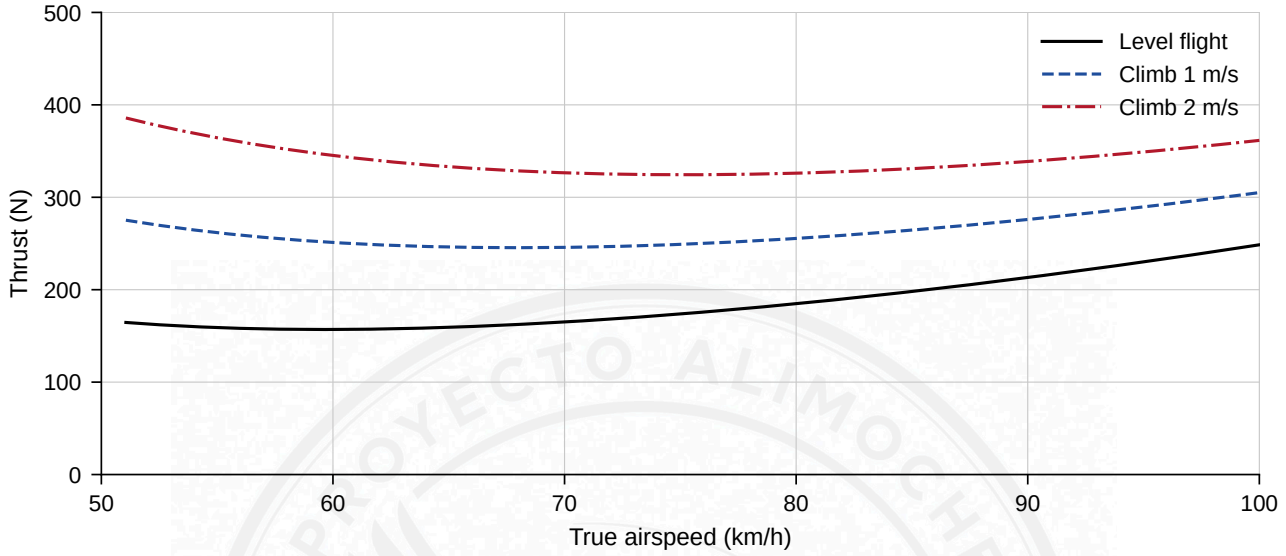


Figure 3. Thrust required with the provisional model at 160 kg and Lillo density.

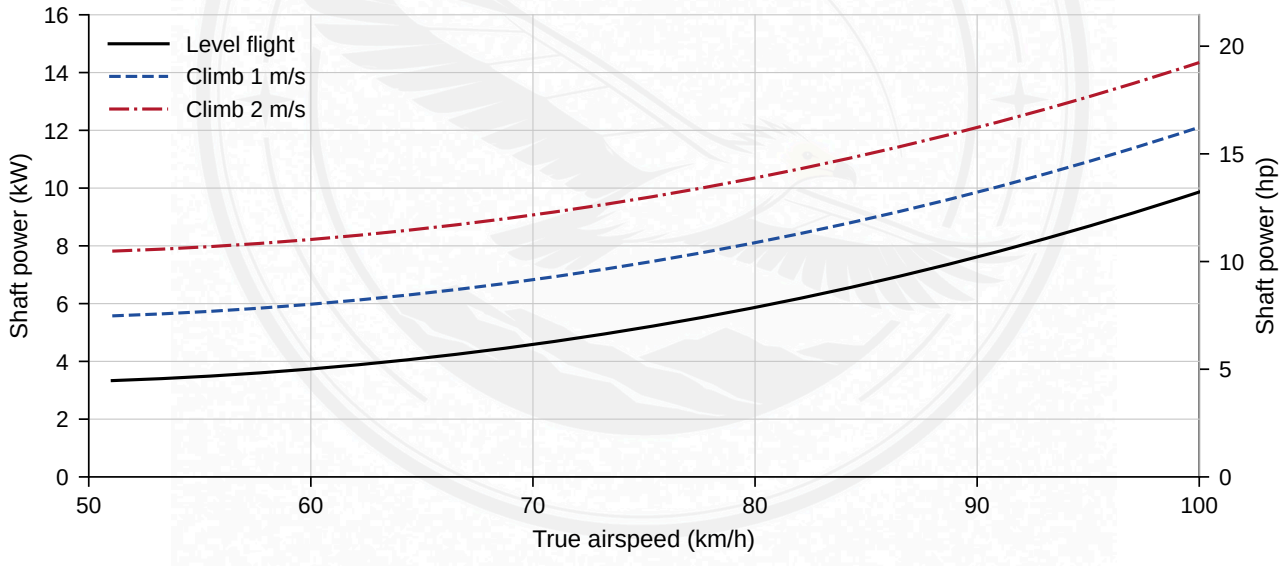


Figure 4. Hypothetical shaft power at a constant propulsive efficiency of 0.70, Lillo, 160 kg. Equivalent scales in kW and mechanical hp (1 hp = 0.7457 kW).

An unpowered glider cannot sustain a climb in still air. The climb charts serve only as input to A05 if propulsion is studied. The engine and its installation would consume mass and increase drag, so the figures are not an engine specification.

6 Sensitivity

Stall speed depends mainly on $C_{L,\max}$ and on wing loading. Glide ratio and power are especially sensitive to the parasite drag of the pilot and the bracing, gathered here in $C_{D,0}$. The factor e also changes the induced drag. The variations in Table 4 change a single parameter at a time from the central case; they show the sensitivity of the model, not uncertainty margins.

Table 4. Sensitivity of the model.

Parameter	Illustrative change	Effect in the 160 kg case
CLmax	1.50 to 1.20	Stall at Lillo: 51.1 to 57.1 km/h (+11.8 %).
CD0	0.055 ± 20 %	Maximum glide ratio: 11.2 with lower CD0; 9.1 with higher CD0.
Factor e	0.70 to 0.60 / 0.80	Maximum glide ratio: 9.3 / 10.7.
Total mass	160 ± 10 kg	Stall at Lillo: 49.4 to 52.6 km/h.
Density	1.225 to 1.04 kg/m³	True airspeeds and optimum sink: +8.5 %; glide ratio unchanged.

The figures highlight the effect of the aerofoil and of the drag of exposed elements. Real geometry changes may alter several parameters at once, so these results cannot be added directly.

7 SG 38 as a configuration reference

The historical SG 38 has approximately 10.4 m span, 6.3 m length and 100–105 kg empty mass, depending on the example [7]. The Deutsches Museum gives a glide ratio of 10 at 52 km/h and a minimum sink of 1.3 m/s at 48 km/h [6]. Its published wing area is about 16 m² [8], against the 10 m² adopted here for ALIMOCHE. The similar span does not allow the polar or the speeds to be transferred without correcting for area, mass and drag. Figure 5 shows the three-view drawing used [3].

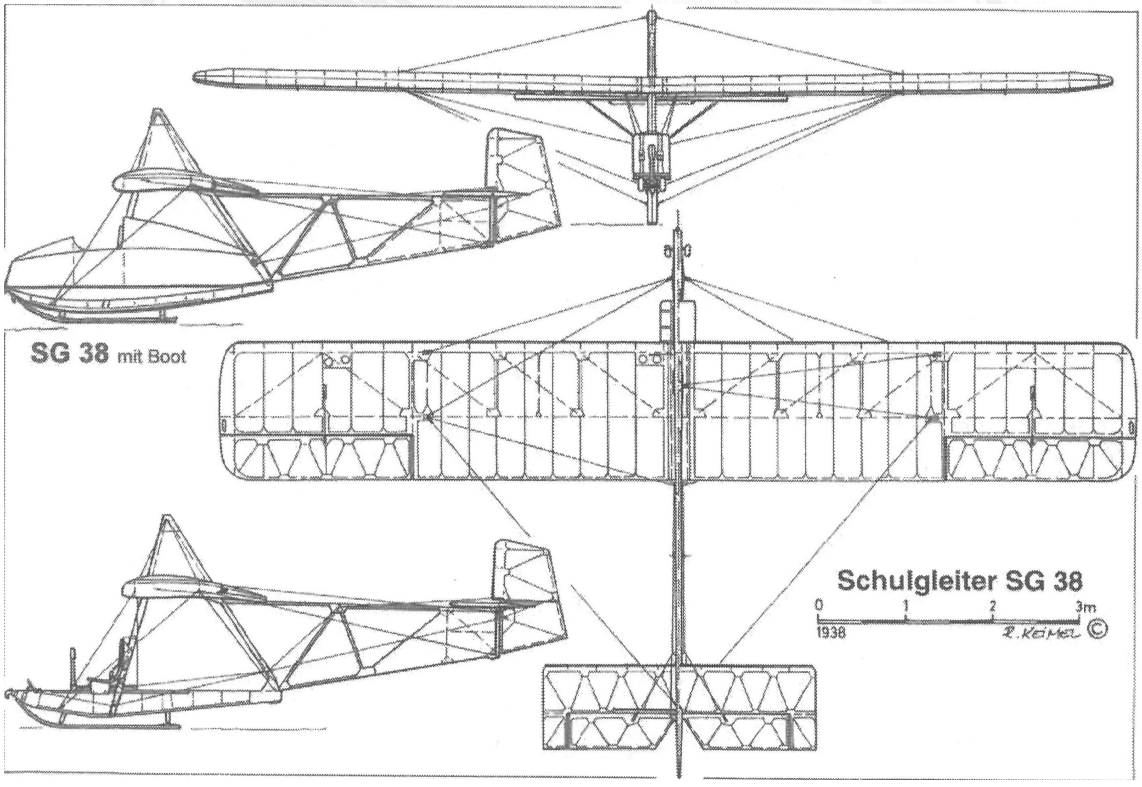


Figure 5. SG 38 three-view sheet provided for this study. The image shows the historical SG 38 and does not define the dimensions of ALIMOCHE. Labels in German in the original drawing: «Schulgleiter» (primary trainer glider), «mit Boot» (with nacelle).

References

[1] ALM-00. *Problem statement and design basis*: empty mass below 70 kg and maximum pilot 90 kg.
[2] ALM-P00. *Specification*: Lillo atmosphere, pilot cases and mission.
[3] SG 38 three-view sheet provided for this study.
[4] NASA Glenn Research Center. Lift equation and induced-drag model. www1.grc.nasa.gov/beginners-guide-to-aeronautics/induced-drag-coefficient/

[5] FAA. *Glider Flying Handbook*, FAA-H-8083-13A, chap. 3: definitions of sink polar and best glide. faa.gov/sites/faa.gov/files/regulations_policies/handbooks_manuals/aviation/glider_handbook/faa-h-8083-13a.pdf

[6] Deutsches Museum. SG 38 data: span, mass, glide ratio and sink. deutsches-museum.de/flugwerft-schleissheim/ausstellung/luftsport-und-zivile-luftfahrt/sg-38

[7] Smithsonian NASM. Length, span and general configuration of the SG 38. airandspace.si.edu/collection-objects/schneider-hofmann-rehberg-sg-38/nasm_A19600311000

[8] Vintage Glider Club. *VGC Newsletter* no. 90, spring 1997, SG 38 data sheet (approximate historical area of 16 m²), PDF.